

- 19** A small diesel engine uses a volume of $1.5 \times 10^4 \text{ cm}^3$ of fuel per hour to produce a useful power output of 40 kW. Assume that 34 kJ of energy is transferred to the engine when it uses 1.0 cm^3 of fuel.

What is the rate of transfer from the engine of energy that is wasted?

- A** 102 kW **B** 142 kW **C** 182 kW **D** 470 kW